

ECE2150J Intro to Circuits

Chapter 9. Sinusoids and Phasors

*Recommend you to bring your calculator to lectures

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9.1 Introduction

- Direct current (dc) vs Alternating current (ac)
- ac is more efficient and economical to transmit over long distances
- Circuits driven by sinusoidal current or voltage sources are called ac circuits.

- A sinusoidal forcing function produces both a transient response and a steady-state response, like the step function in Chapters 7 and 8. we say that the circuit is operating at sinusoidal steady state.
- We are interested in sinusoidal **steady-state** response of AC circuits.

9.2 Sinusoids

A sinusoid is a signal that has the form of the sine or cosine function.

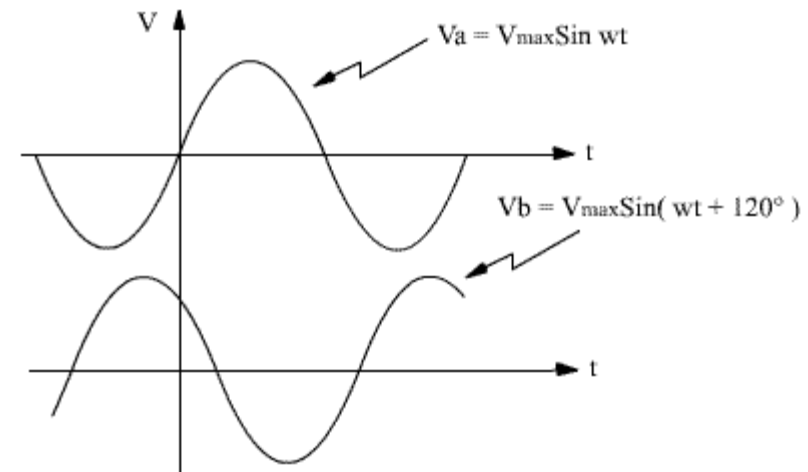
$$v(t) = V_m \sin(\omega t + \phi)$$

V_m : amplitude

ω : angular frequency

ϕ : initial phase

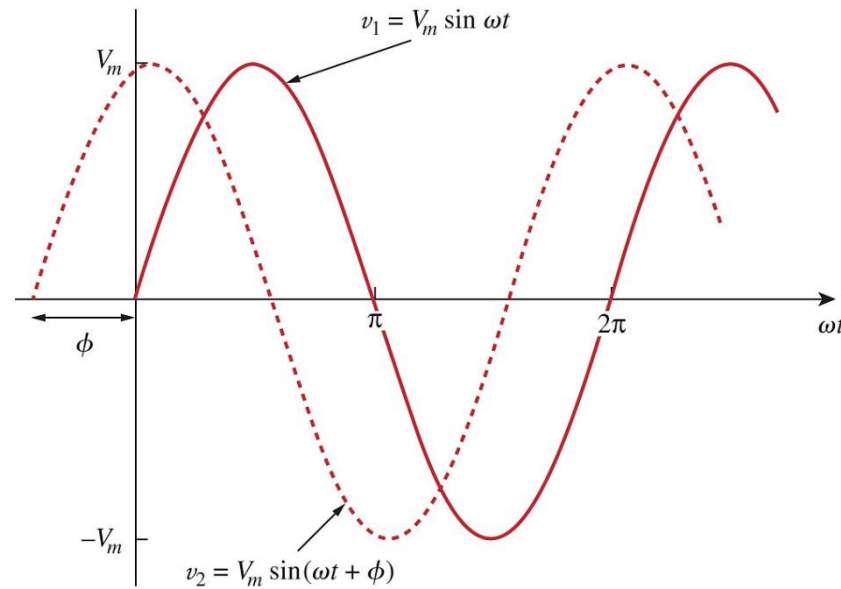
ωt : argument of the sinusoid



e.g. we have two sinusoids

$$v_1(t) = V_m \sin \omega t$$

$$v_2(t) = V_m \sin(\omega t + \phi)$$



The starting point of v_2 occurs first in time. Therefore, we say that v_2 **leads** v_1 by ϕ or that v_1 **lags** v_2 by ϕ . If $\phi \neq 0$, we say that v_1 and v_2 are **out of phase**. If $\phi = 0$, then v_1 and v_2 are said to be **in phase**.

9.3 Phasors

Sinusoids are easily expressed in terms of **phasors**, which are more convenient to work with than sine and cosine functions. A phasor is a **complex number** that represents **the amplitude and phase of a sinusoid**.

*Complex Numbers:

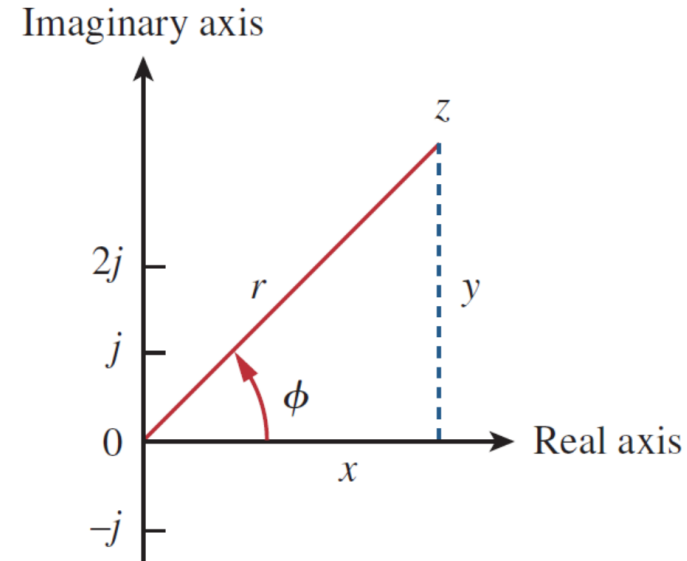
- $z = x + jy$ Rectangular form

$$x = r \cos \phi, \quad y = r \sin \phi$$

- $z = r \angle \Phi$ Polar form

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}$$

- $z = r e^{j\Phi}$ Exponential form



*Basic properties of complex numbers

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \angle -\phi \quad \frac{1}{j} = -j$$

Square Root:

$$\sqrt{z} = \sqrt{r} \angle \phi/2$$

Complex Conjugate:

$$z^* = x - jy = r \angle -\phi = re^{-j\phi}$$

Given a sinusoid

$$v(t) = V_m \cos(\omega t + \phi)$$

$$= \text{Re}\left(V_m e^{j(\omega t + \phi)}\right)$$

$$= \text{Re}\left(V_m e^{j\phi} e^{j\omega t}\right)$$

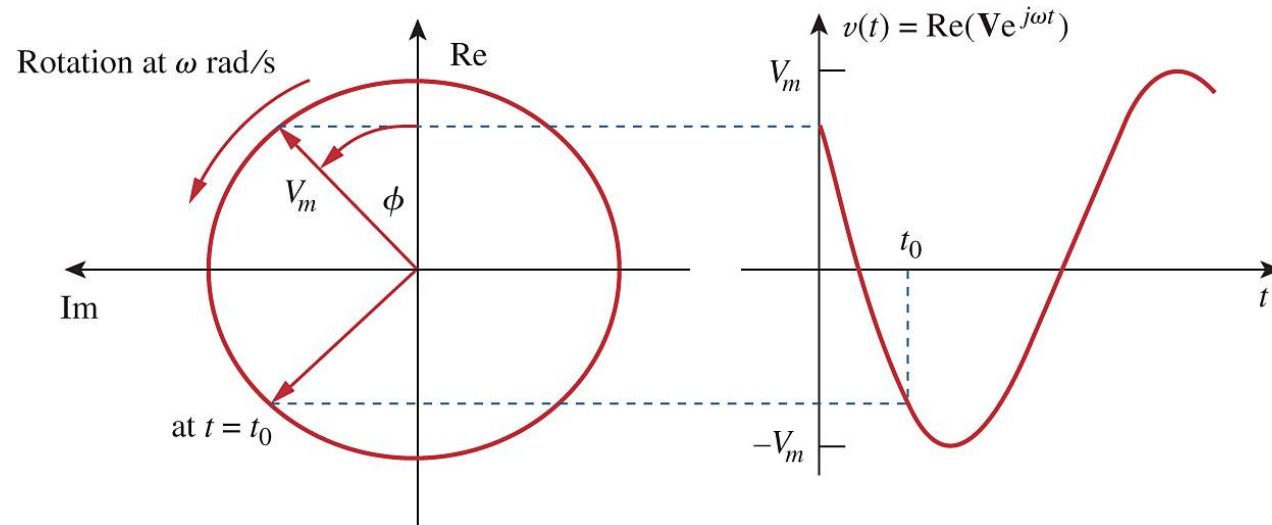
$$= \text{Re}(\tilde{V} e^{j\omega t}) \quad \text{where} \quad \tilde{V} = V_m e^{j\phi} = V_m \angle \phi$$

***Euler's identity**

$$e^{\pm j\phi} = \cos\phi \pm j\sin\phi$$

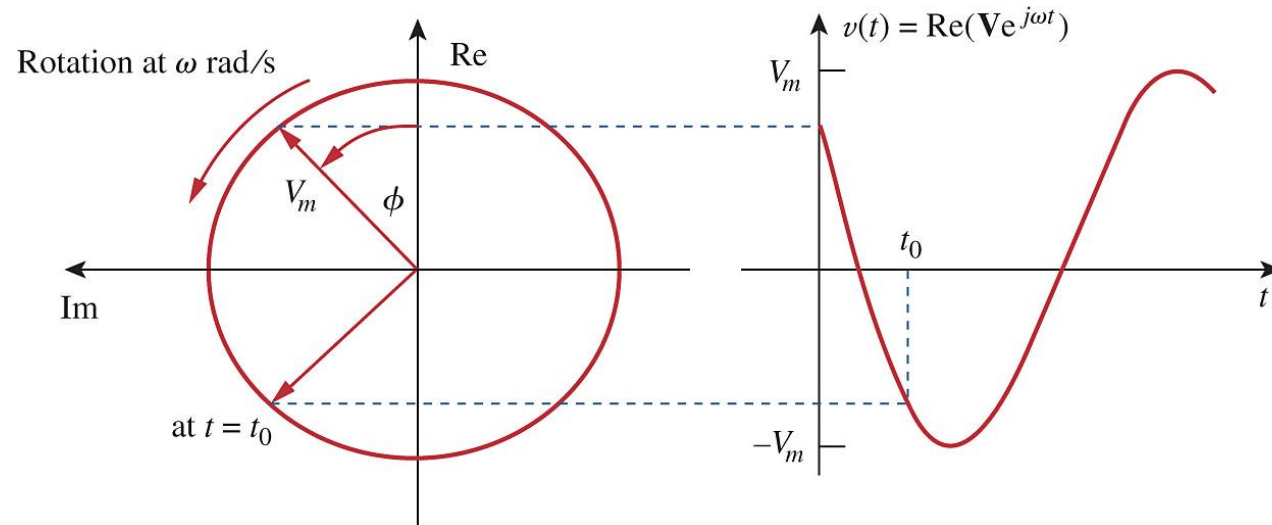
- $\cos\phi = \text{Re}(e^{j\phi})$
- $\sin\phi = \text{Im}(e^{j\phi})$

Phasor representation of the sinusoid $v(t)$ $\tilde{V} = V_m e^{j\phi} = V_m \angle \phi$



The sinor $\tilde{V}e^{j\omega t} = V_m e^{j(\omega t + \phi)}$ on the complex plane

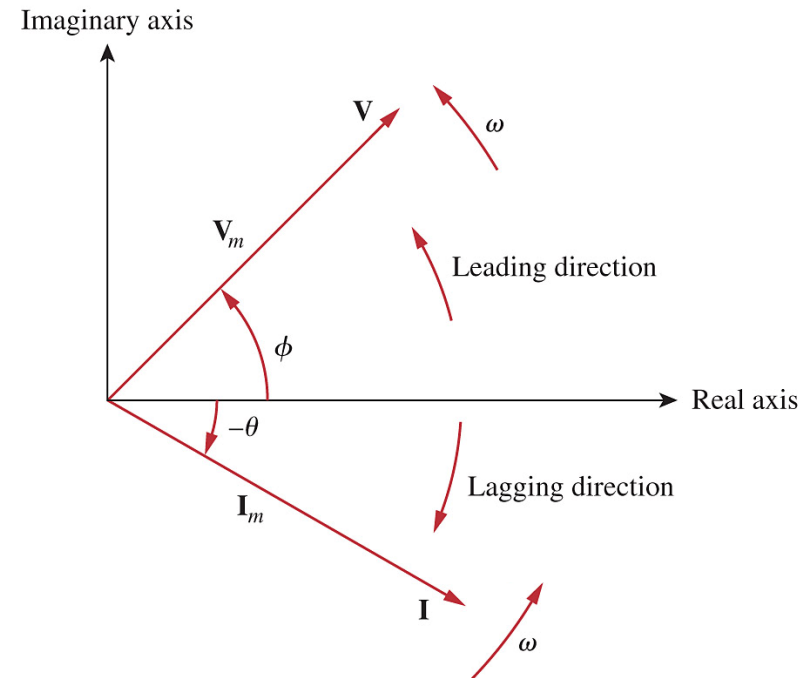
As time increases, the sinor rotates on a circle of **radius V_m at an angular velocity ω in the counter clockwise direction**. We may regard $v(t)$ as the projection of the sinor on the real axis.



The sinor $\tilde{V}e^{j\omega t} = V_m e^{j(\omega t + \phi)}$ on the complex plane

The value of the sinor **at time $t=0$** is the phasor **$\mathbf{V}$** . The sinor may be regarded as a rotating phasor. Thus, whenever a sinusoid is expressed as a phasor, **the term $e^{j\omega t}$ is implicitly present.**

Because a phasor has **magnitude** and **phase** (“direction”), it behaves as a vector.



$$\tilde{V} = V_m \angle \phi$$

$$\tilde{I} = I_m \angle -\theta.$$

Phasor diagram: graphical representation of phasors

A sinusoid has a time-domain representation $v(t) = V_m \cos(\omega t + \phi)$ and a phasor-domain representation $\tilde{V} = V_m \angle \phi$.

The phasor domain is also known as the frequency domain.

$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow \tilde{V} = V_m \angle \phi$$

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

Example 9.3

$$(b) \frac{10 \angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*}$$

Example 9.3

$$(b) \frac{10 \angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*} = 0.565 \angle -160.13$$

$$10\cos(-30) + j10\sin(-30) = 8.66 - j5$$

$$8.66 - j5 + 3 - j4 = 11.66 - j9$$

$$(2 + j4)(3 - j5)^* = (2 + j4)(3 + j5) = -14 + j22$$

$$11.66 - j9 \rightarrow \sqrt{11.66^2 + 9^2} = 14.73$$

$$-14 + j22 \rightarrow \sqrt{14^2 + 22^2} = 26.07$$

$$\text{Angle: } \tan^{-1}\left(\frac{-9}{11.66}\right) = -37.66$$

$$\text{Angle: } \tan^{-1}\left(\frac{22}{-14}\right) = -57.53 ??$$

Practice Problem 9.3

$$(b) \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5 = 8.29 + j7.20$$

Practice Problem 9.3

$$(b) \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5 = 8.29 + j7.20$$

$$10 + j5 + 2.30 + j1.93 = 12.30 + j6.93$$

$$12.30 + j6.93 \rightarrow \sqrt{12.30^2 + 6.93^2} = 14.12$$

$$-3 + j4 \rightarrow \sqrt{3^2 + 4^2} = 5$$

$$\text{Angle: } \tan^{-1} \left(\frac{6.93}{12.30} \right) = 29.40$$

$$\text{Angle: } \tan^{-1} \left(\frac{4}{-3} \right) = 126.87$$

$$2.82 \text{ angle } -97.47$$

9.4 Phasor Relationships for Circuit Elements

Let's think about the IV **at the resistor**. If the current through a resistor R is $i = I_m \cos(\omega t + \phi)$

The voltage across it is given by Ohm's law as

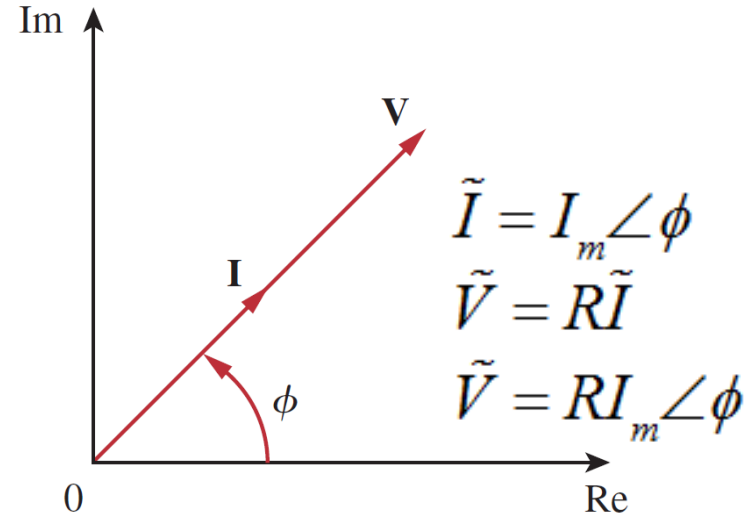
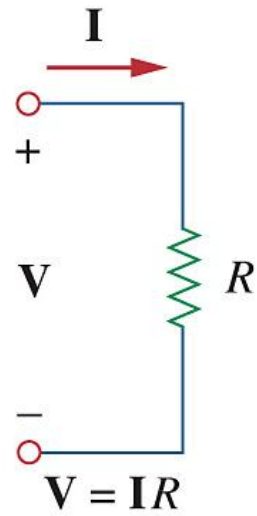
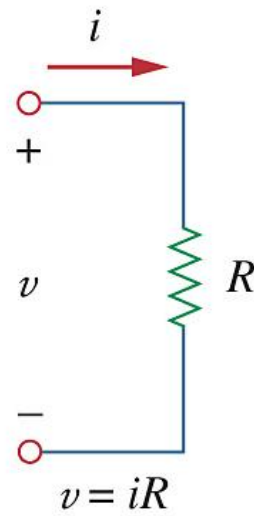
$$v = iR = RI_m \cos(\omega t + \phi)$$

The phasor representation of the voltage is

$$\tilde{V} = RI_m \angle \phi$$

And **the phasor** representation of the current is

$$\tilde{I} = I_m \angle \phi.$$



The voltage-current relation for the resistor in the phasor domain.

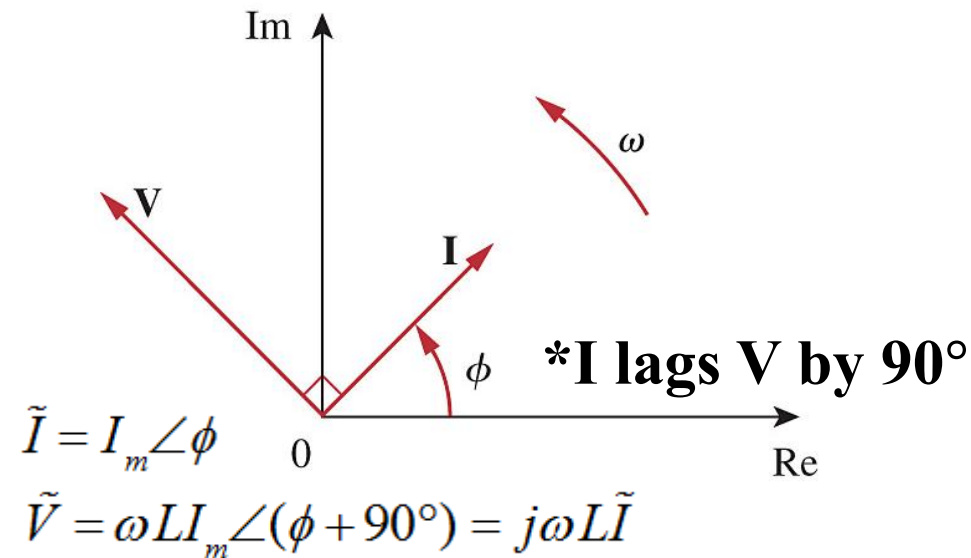
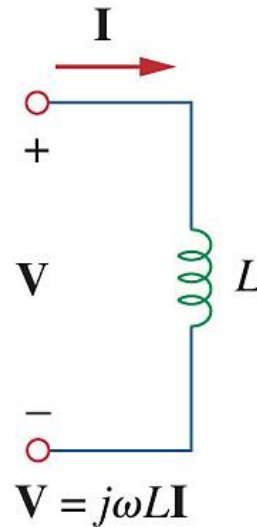
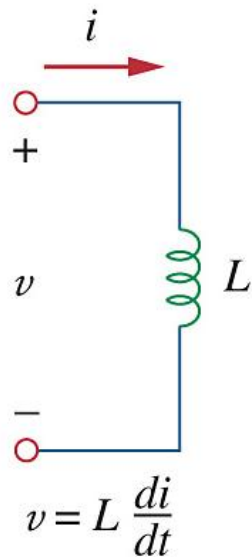
Inductor

For the inductor L , if $i = I_m \cos(\omega t + \phi) \Leftrightarrow \tilde{I} = I_m \angle \phi$
 $v = L \frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi) = \omega L I_m \cos(\omega t + \phi + 90^\circ)$

$$* -\sin(\omega t + \phi) = \cos(\omega t + \phi + 90^\circ)$$

$$\Rightarrow \tilde{V} = \omega L I_m \angle(\phi + 90^\circ) = j\omega L I_m \angle \phi = j\omega L \tilde{I}$$

$$\angle 90^\circ = e^{j90^\circ} = j$$



Capacitor

For the capacitor C , if $v = V_m \cos(\omega t + \phi) \Leftrightarrow \tilde{V} = V_m \angle \phi$

$$i = C \frac{dv}{dt} = -\omega C V_m \sin(\omega t + \phi) = \omega C V_m \cos(\omega t + \phi + 90^\circ)$$

$$\Rightarrow \tilde{I} = \omega C V_m \angle(\phi + 90^\circ) = j\omega C V_m \angle \phi = j\omega C \tilde{V}$$

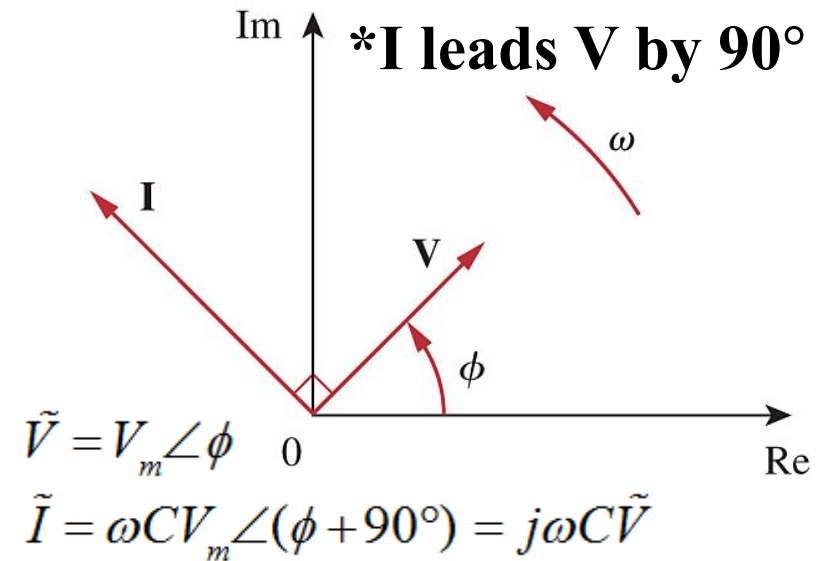
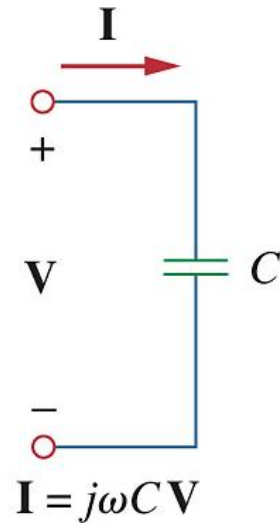
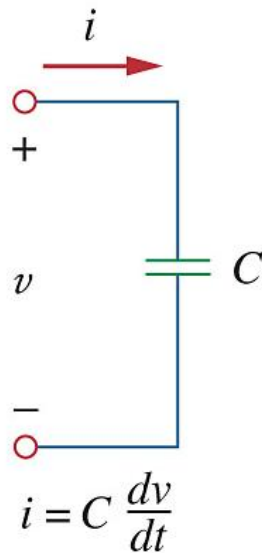
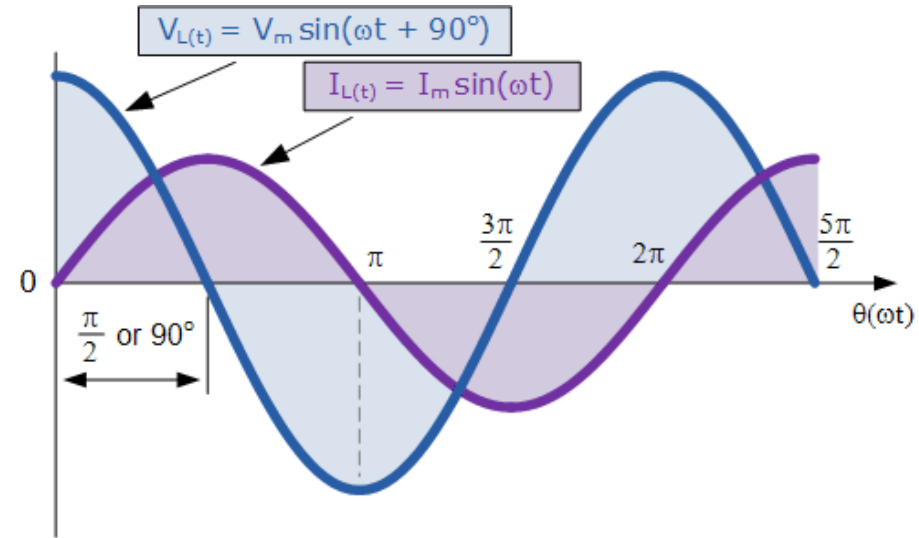
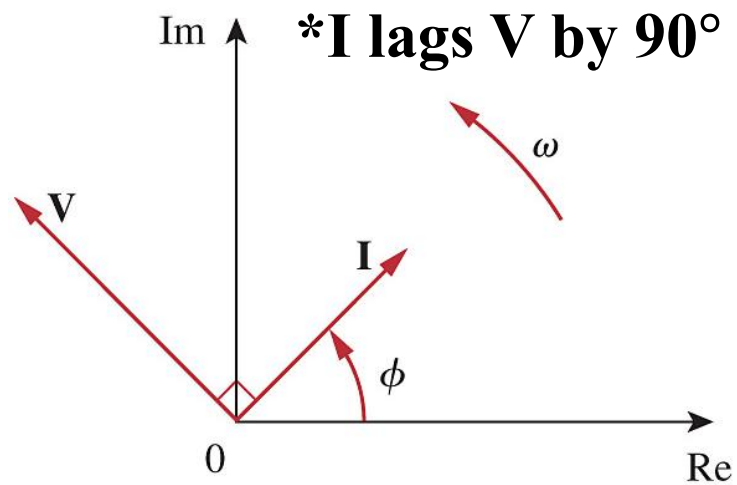


TABLE 9.2

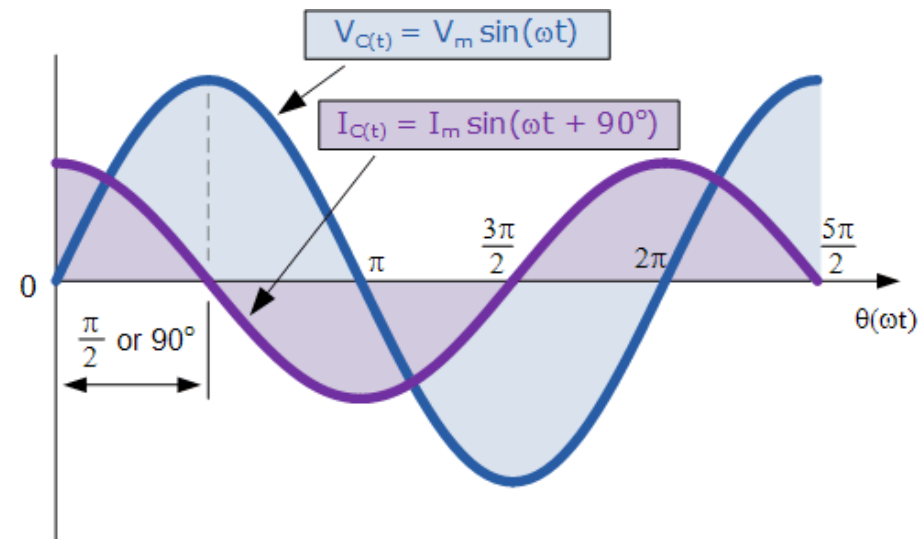
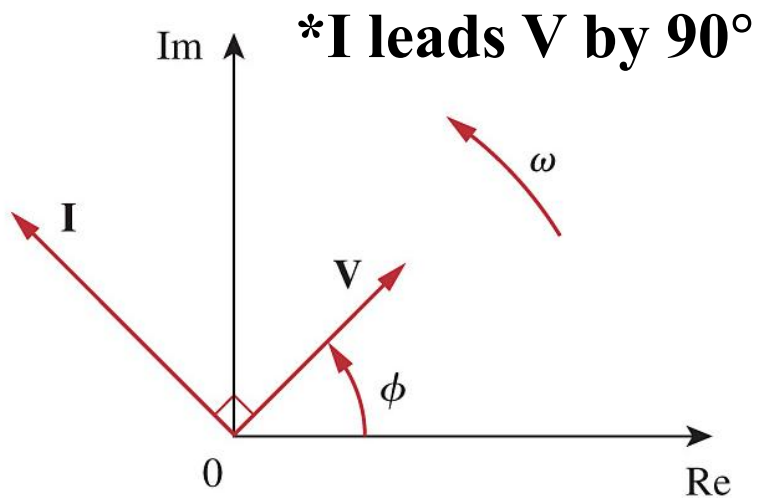
Summary of voltage - current relationships

Element	Time domain	Frequency domain	
R	$v = Ri$	$\tilde{V} = R\tilde{I}$	No delay
L	$v = L \frac{di}{dt}$	$\tilde{V} = j\omega L\tilde{I}$	V faster
C	$i = C \frac{dv}{dt}$	$\tilde{V} = \frac{1}{j\omega C}\tilde{I}$	I faster

Inductor



Capacitor



9.5 Impedance and Admittance

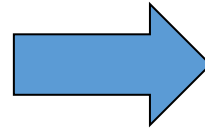
General definition of R

R: Resistance

G: Conductance

$$R = 1/G$$

(Real number)



Z: Impedance

Y: Admittance

$$Z = 1/Y$$

(Complex number)

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

Impedance

The impedance **Z** of a circuit is the ratio of the phasor voltage **V** to the phasor current **I**, measured in Ohms (Ω)

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}$$

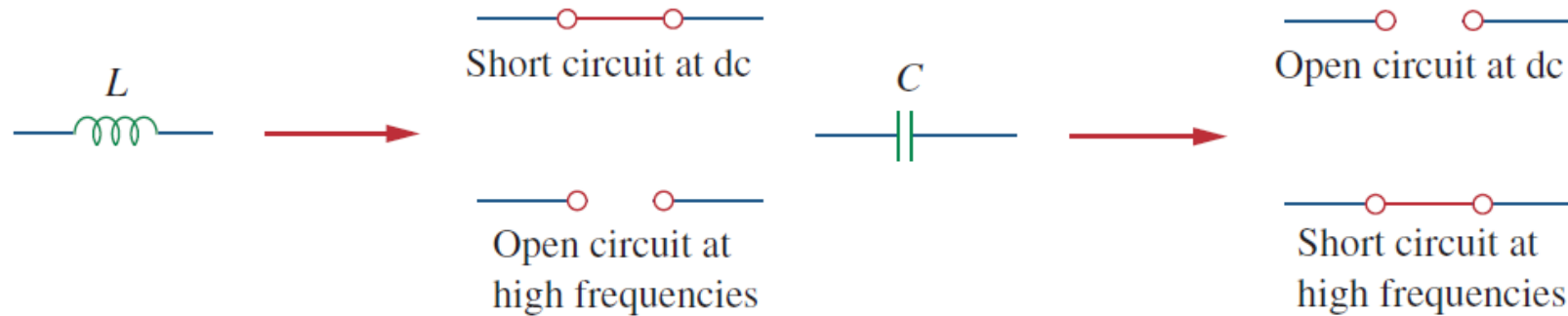
Resistance	$\tilde{V} = R\tilde{I}$	$Z = R$
Inductor	$\tilde{V} = j\omega L\tilde{I}$	$Z = j\omega L$
Capacitor	$\tilde{V} = \frac{1}{j\omega C}\tilde{I}$	$Z = 1 / j\omega C$

Inductor $\tilde{V} = j\omega L \tilde{I} \quad Z = j\omega L$

Capacitor $\tilde{V} = \frac{1}{j\omega C} \tilde{I} \quad Z = 1 / j\omega C$

(i) $\omega = 0$ (DC source): $Z_L = 0, Z_C \rightarrow \infty$

(ii) $\omega \rightarrow \infty$: $Z_L \rightarrow \infty, Z_C = 0$



As a complex quantity, the impedance may be expressed in rectangular form or polar form

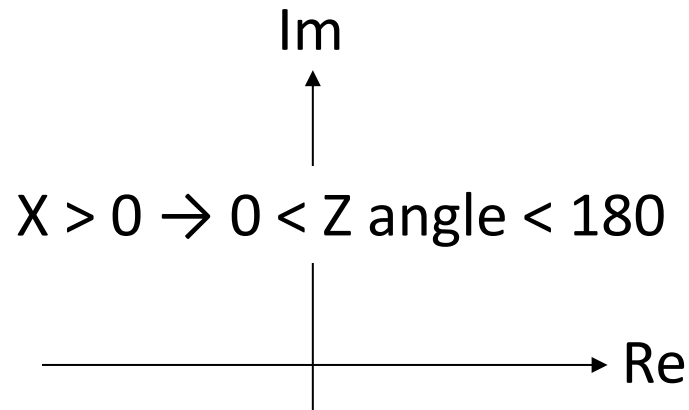
$$Z = R + jX = |Z| \angle \theta \quad |Z| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

where

$$R : \text{resistance} \quad R = |Z| \cos \theta$$

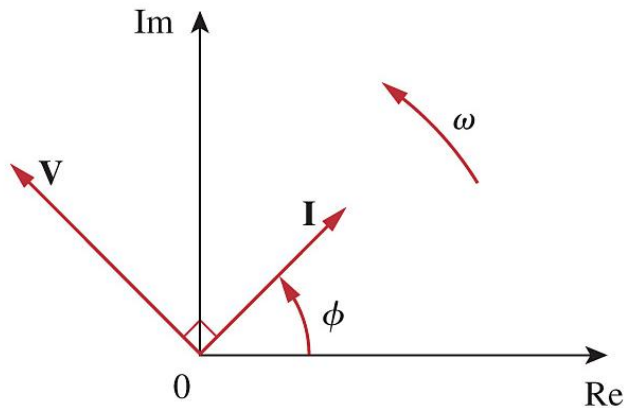
$$X : \text{reactance} \quad X = |Z| \sin \theta$$

i) $X > 0$, the impedance is **inductive or lagging**, i.e. current lags voltage.



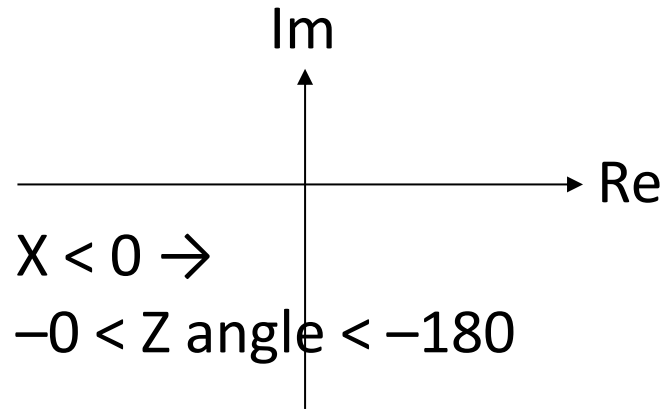
$$V = IZ$$

$$\text{Angle (V)} = \text{Angle (I)} + \text{Angle (Z)}$$



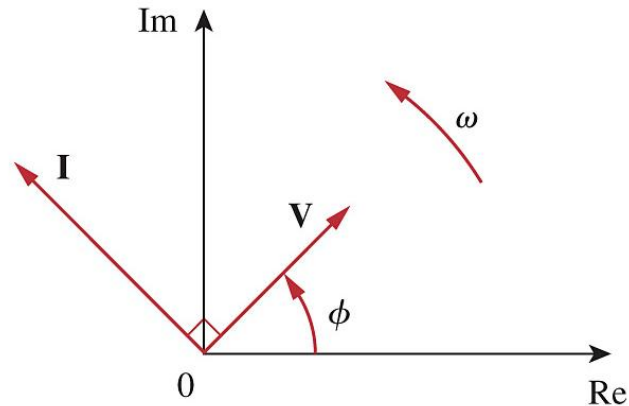
$Z = R + jX$ where $X > 0$
 $\rightarrow$ Impedance is **inductive**
or **lagging** (I lags V)

ii) $X < 0$, the impedance is **capacitive or leading**, i.e. current leads voltage.



$$I = \frac{V}{Z}$$

Angle (I) = Angle (V) – Angle (Z)
where Angle (Z) is negative



$Z = R + jX$ where $X < 0$
 $\rightarrow$ Impedance is **capacitive**
or **leading** (I leads V)

Admittance

The admittance **Y** of a circuit is the ratio of the phasor current **I** to the phasor voltage **V**, measured in siemens (S).

$$Y = \frac{I}{V} = \frac{1}{Z}$$

Resistance	$\tilde{V} = R\tilde{I}$	$Y = 1 / R$
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Inductor	$\tilde{V} = j\omega L\tilde{I}$	$Y = 1 / j\omega L$
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Capacitor	$\tilde{V} = \frac{1}{j\omega C}\tilde{I}$	$Y = j\omega C$
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The admittance (Y) can be written as

$$G + jB = \frac{1}{R + jX}$$

where

G : conductance

B : susceptance

The admittance, conductance, and
susceptance are all measured in siemens.

9.6 Kirchhoff's Laws in the Frequency Domain

For KVL, let $v_1, v_2, \dots, v_n$ be the voltages around a closed loop.

Then,
$$\sum_{i=1}^n v_i = 0$$

In sinusoidal steady state,

$$v_i = V_{mi} \cos(\omega t + \phi_i) = \operatorname{Re}(\tilde{V}_i e^{j\omega t})$$

$$\sum_{i=1}^n v_i = 0 \quad \leftarrow \quad v_i = V_{mi} \cos(\omega t + \phi_i) = \operatorname{Re}(\tilde{V}_i e^{j\omega t})$$

$$\Rightarrow \sum_{i=1}^n \operatorname{Re}(\tilde{V}_i e^{j\omega t}) = 0$$

$$\operatorname{Re}\left(\sum_{i=1}^n \tilde{V}_i\right) e^{j\omega t} = 0$$

but $e^{j\omega t} \neq 0$,

$$\sum_{i=1}^n \tilde{V}_i = 0$$

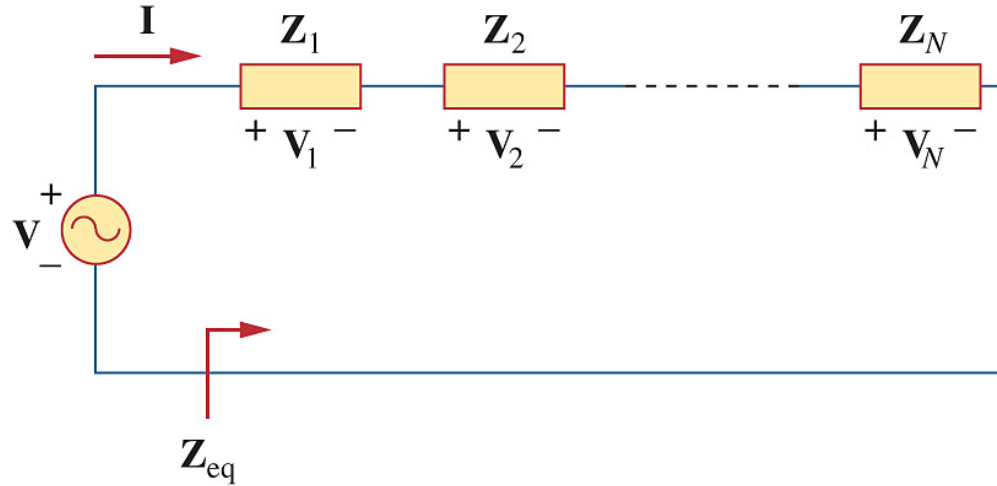
indicating that KVL holds for phasors.

For KCL, if $i_1, i_2, \dots, i_n$ are the currents leaving or entering a closed surface in a circuit at time t , and $I_1, I_2, \dots, I_n$ are the phasor forms of $i_1, i_2, \dots, i_n$,

$$\text{Then, } \sum_{i=1}^n i_i = 0 \Rightarrow \sum_{i=1}^n \tilde{I}_i = 0$$

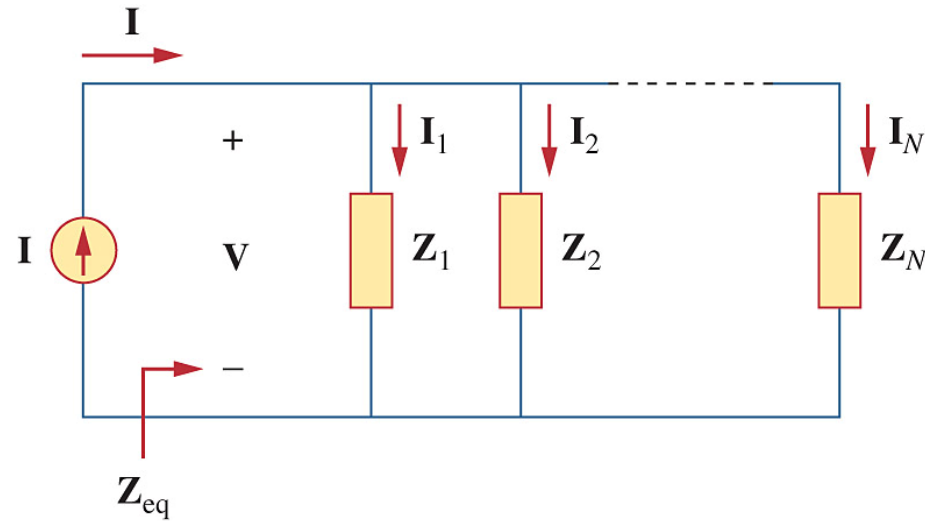
Since basic circuit laws, **Kirchoff's and Ohm's, hold in phasor domain**, it is not difficult to analyze ac circuit.

9.7 Impedance Combinations



For the N series-connected impedances, the equivalent impedance at the input terminal is

$$Z_{eq} = \frac{\tilde{V}}{\tilde{I}} = \frac{\sum_{i=1}^N \tilde{V}_i}{\tilde{I}} = \sum_{i=1}^N \frac{\tilde{V}_i}{\tilde{I}} = \sum_{i=1}^N Z_i$$

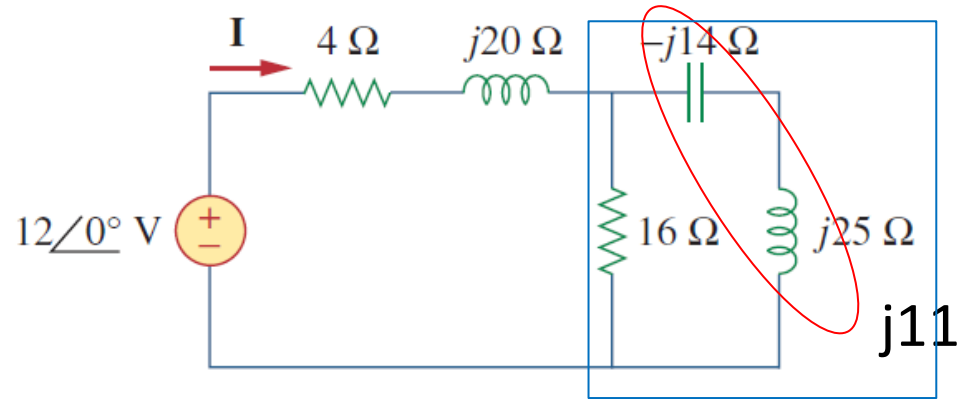


For the N parallel-connected impedances, the equivalent admittance at the input terminal is

$$Y_{eq} = \frac{\tilde{I}}{\tilde{V}} = \frac{\sum_{i=1}^N \tilde{I}_i}{\tilde{V}} = \sum_{i=1}^N \frac{\tilde{I}_i}{\tilde{V}} = \sum_{i=1}^N Y_i$$

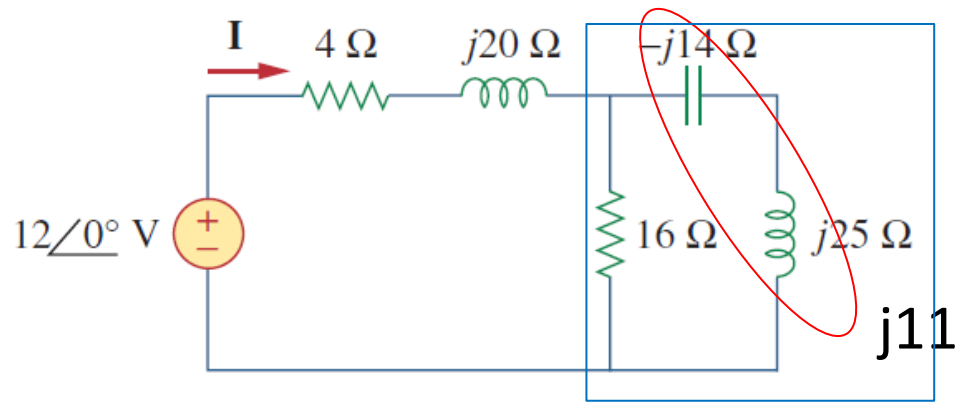
Example

9.39 For the circuit shown in Fig. 9.46, find Z_{eq} and use that to find current $\mathbf{I}$. Let $\omega = 10$ rad/s.



Example

9.39 For the circuit shown in Fig. 9.46, find Z_{eq} and use that to find current $\mathbf{I}$. Let $\omega = 10$ rad/s.



$$16 \parallel j11 = \frac{j176}{16 + j11} = \frac{j176}{19.42 \angle 34.51} = 9.06 \angle 55.49$$

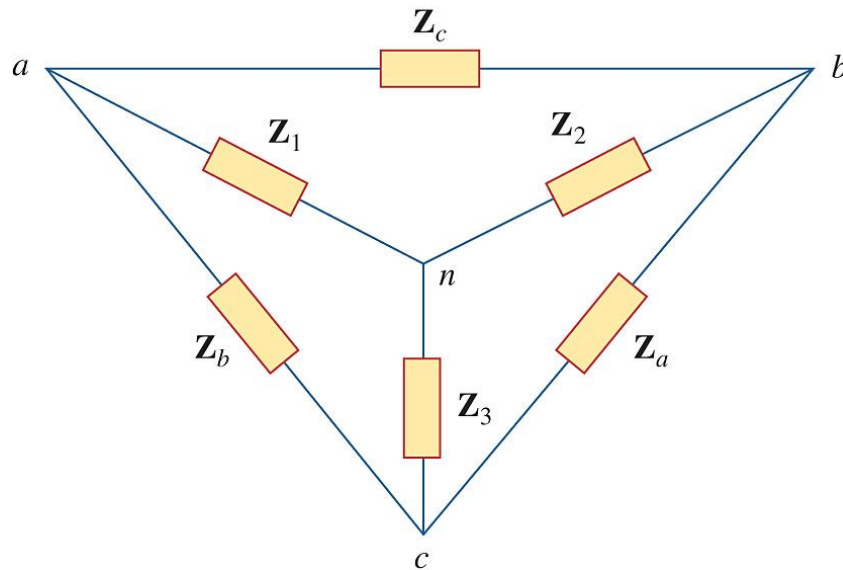
$$\begin{aligned} Z_{eq} &= 9.06 \angle 55.49 + 4 + j20 = 5.13 + j7.47 + 4 + j20 \\ &= \mathbf{9.13 + j27.47} \end{aligned}$$

$$I = \frac{V}{Z_{eq}} = \frac{12 \angle 0}{9.13 + j27.47} = \frac{12 \angle 0}{28.95 \angle 71.62} = 0.41 \angle (-71.62)$$

$$\mathbf{i(t) = 0.41 \cos(10t - 71.62) [A]}$$

The **delta-to-wye** and **wye-to-delta** transformations that we applied to resistive circuits are also valid for impedances.

(i) Y-Delta



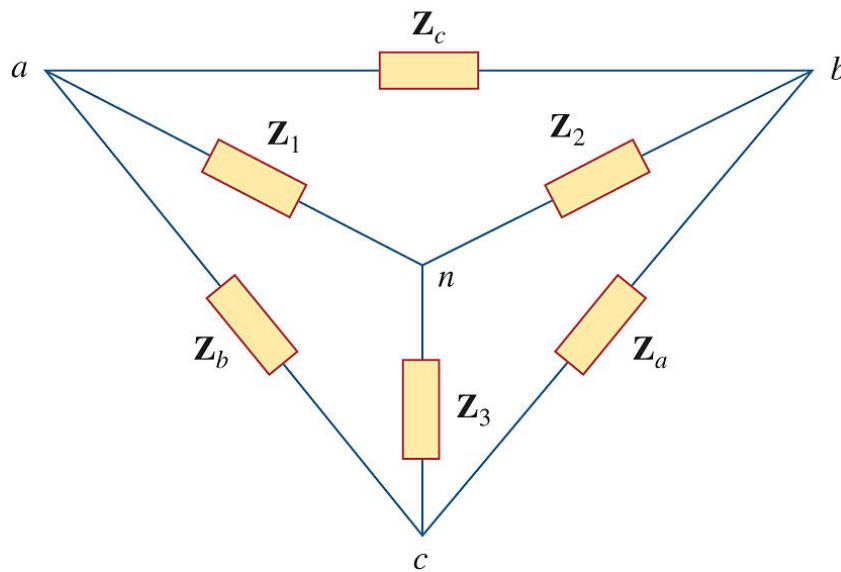
Y - Δ conversion:

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

(ii) Delta-Y



Δ -Y conversion:

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$

Practice Problem 9.10 Find the input impedance of the circuit in Fig. 9.24 at $\omega = 10 \text{ rad/s}$.

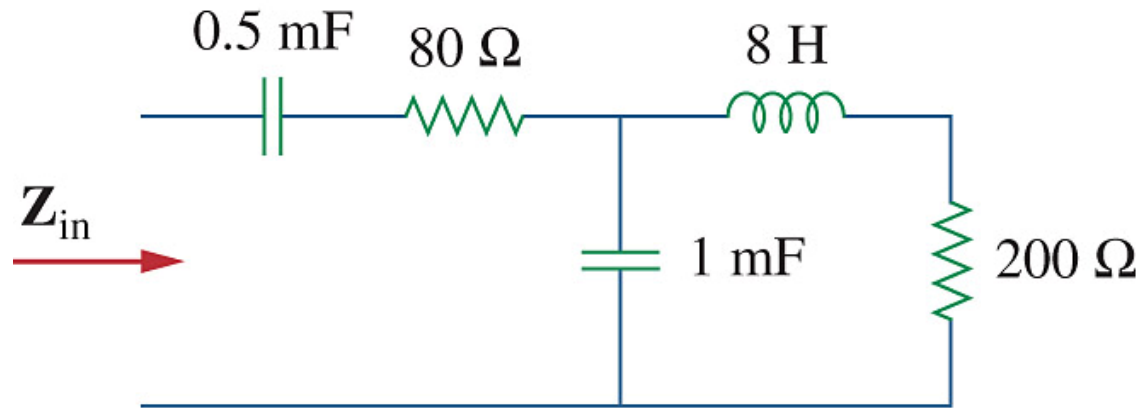
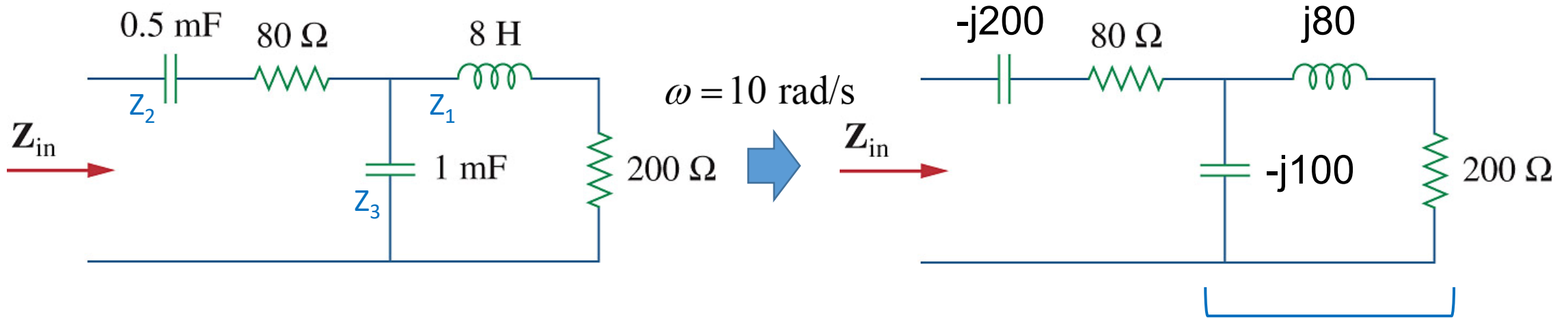


Figure 9.24



Solution :

8-H inductor: $Z_1 = j10 \times 8 = j80 \text{ } (\Omega)$

0.5-mF capacitor: $Z_2 = \frac{1}{j10 \times (0.5 \times 10^{-3})}$

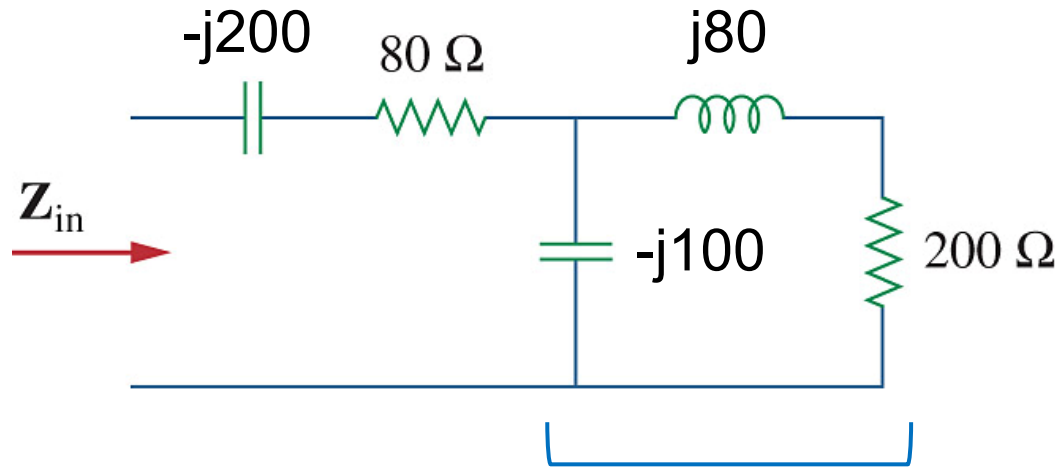
$= -j200 \text{ } (\Omega)$

1-mF capacitor: $Z_3 = \frac{1}{j10 \times (1 \times 10^{-3})}$

$= -j100 \text{ } (\Omega)$

$$= \frac{(-j100) \times (j80 + 200)}{(-j100) + (j80 + 200)} \approx 107.1688 \angle -62.49^\circ$$

$$\approx 49.5016 - j95.0512 \text{ } (\Omega)$$



$$= \frac{(-j100) \times (j80 + 200)}{(-j100) + (j80 + 200)} \approx 107.1688 \angle -62.49^\circ$$

$$\approx 49.5016 - j95.0512\ (\Omega)$$

$$Z_{in} = -j200 + 80 + 49.5016 - j95.0512$$

$$\approx 129.50 - j295.05\ (\Omega)$$

Practice Problem 9.11 Calculate v_o in the circuit of Fig. 9.27.

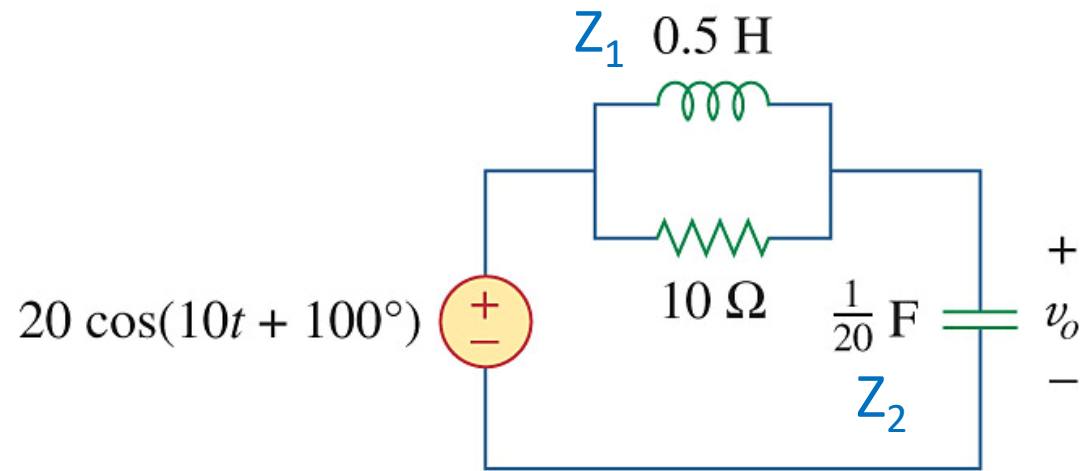
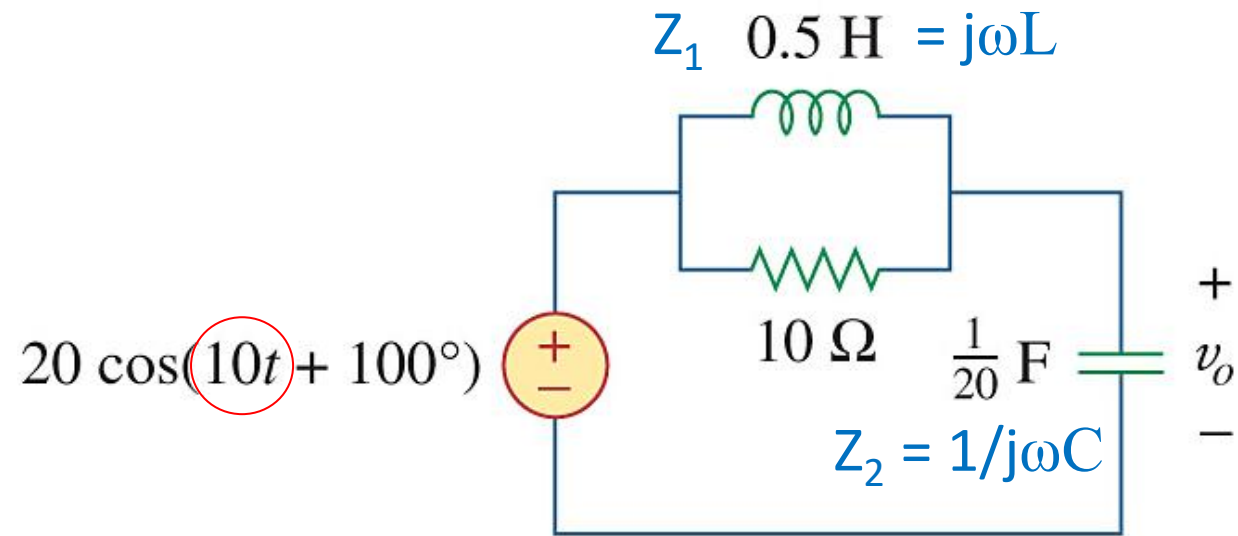


Figure 9.27

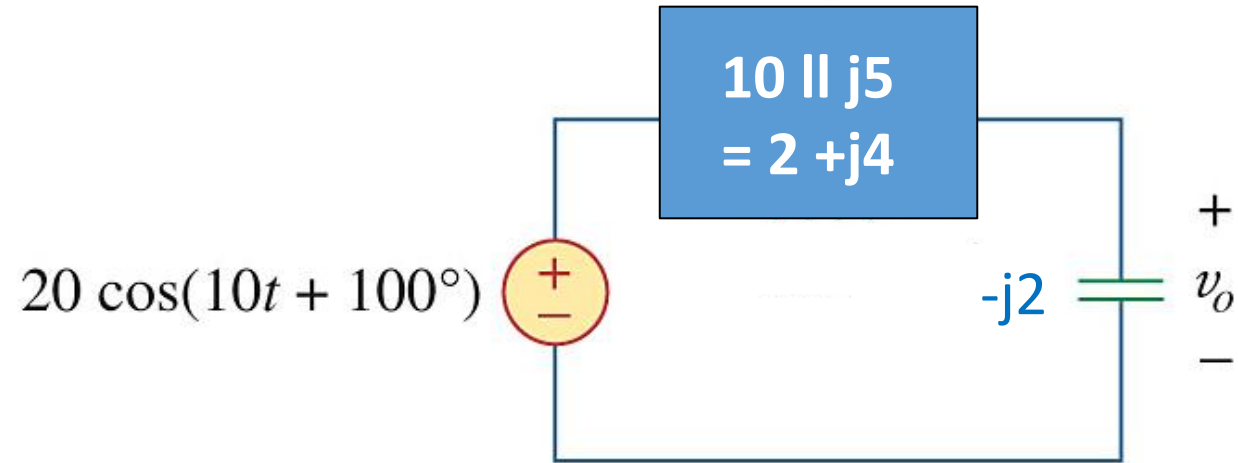


$$\omega = 10, \mathbf{V} = 20\angle 100$$

$$0.5\text{-H inductor: } Z_1 = j10 \times 0.5 = j5 \, (\Omega)$$

$$\frac{1}{20}\text{-F capacitor: } Z_2 = \frac{1}{j10 \times (1/20)}$$

$$= -j2 \, (\Omega)$$



$$\tilde{V}_o = 20 \angle 100^\circ \times \frac{-j2}{-j2 + 10 \parallel j5}$$

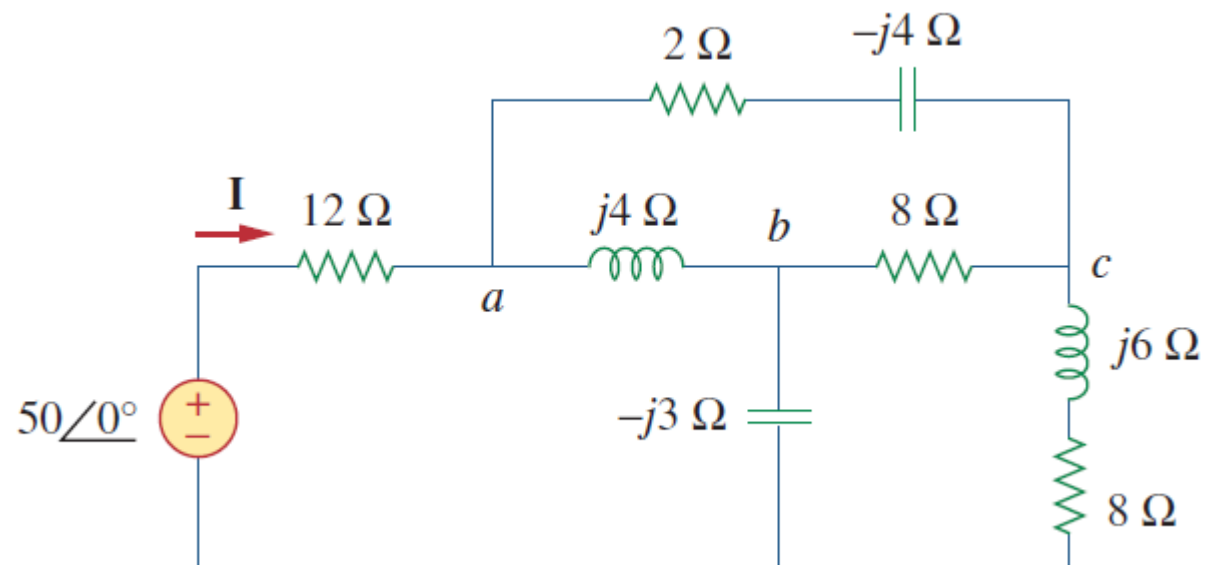
$$10 \parallel j5 = \frac{10 \times j5}{10 + j5} = \frac{j10}{2 + j} = 2 + j4 \quad \text{Or } 10 \parallel j5 = 4.47 \angle 63.43^\circ$$

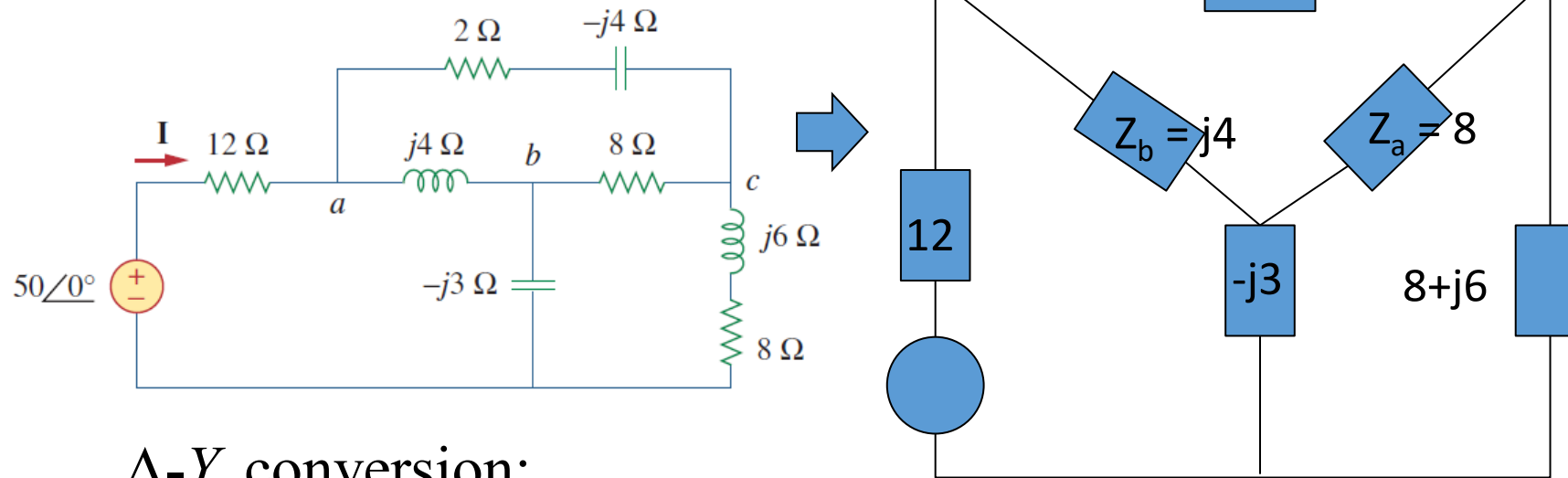
$$\tilde{V}_o = 20 \angle 100^\circ \times \frac{-j2}{2 + j2} = 10\sqrt{2} \angle -35^\circ \text{ (V)}$$

$$v_o(t) = 10\sqrt{2} \cos(10t - 35^\circ) \text{ (V)}$$

Example 9.12

Find current $\mathbf{I}$ in the circuit of Fig. 9.28.



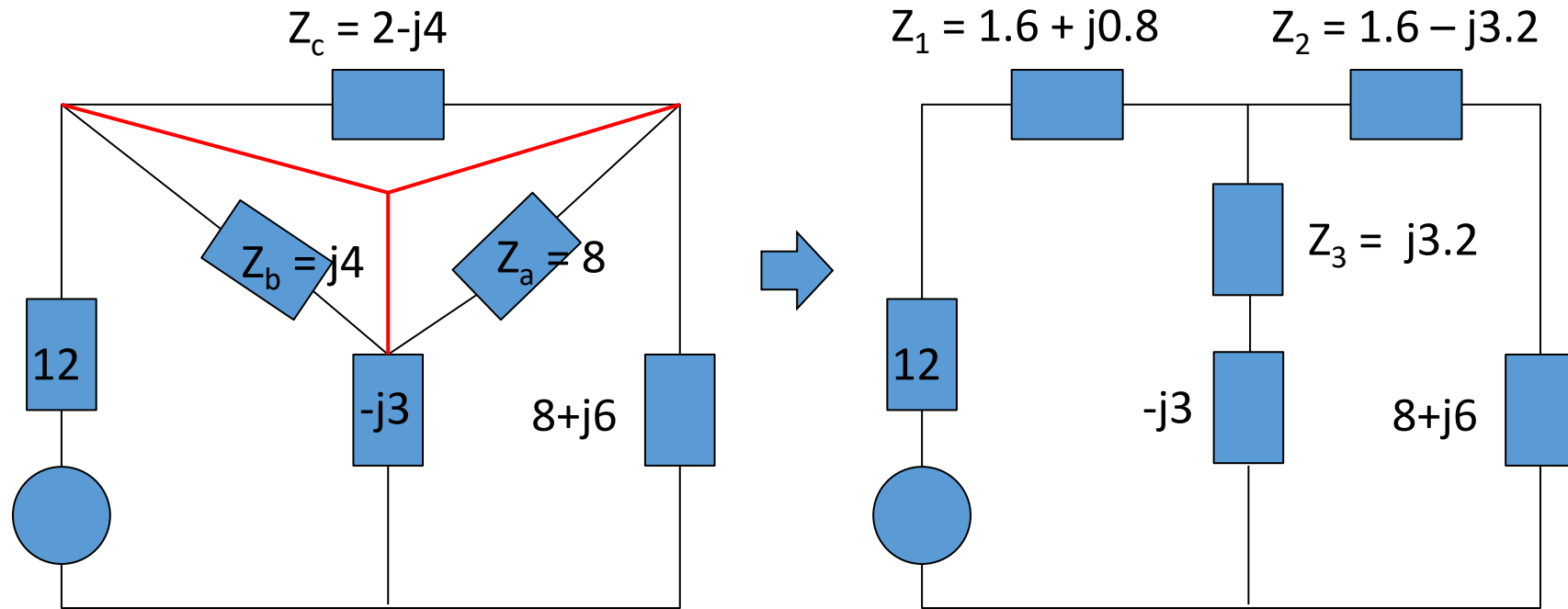


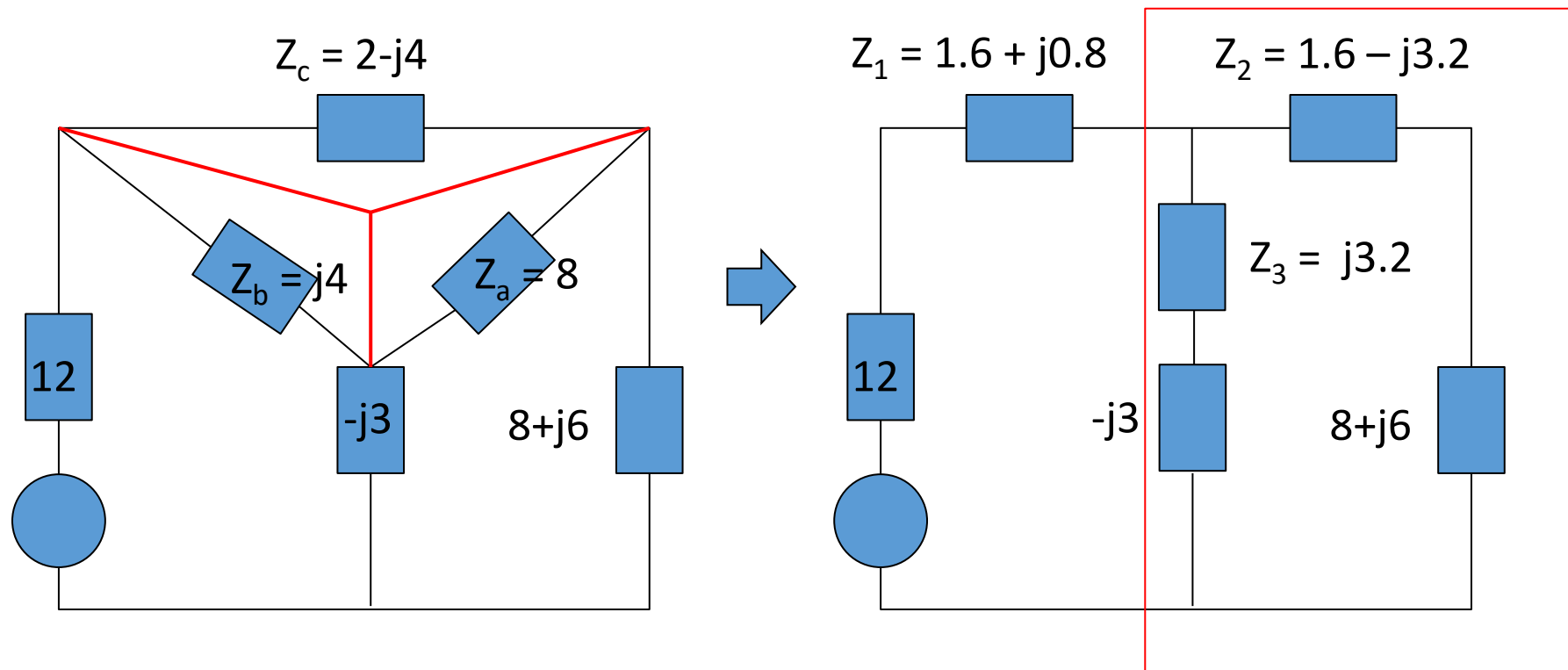
Δ -Y conversion:

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c} = 1.6 + j0.8$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c} = 1.6 - j3.2$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c} = j3.2$$





$$j0.2 \parallel 9.6 + j2.8$$

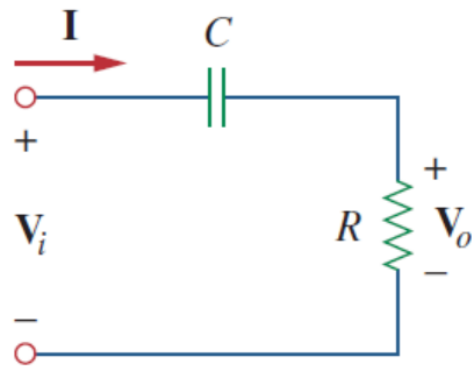
$$= 0.2 \text{ angle } (88.91)$$

$$Z_{eq} = 13.6 + j0.8 + 0.004 + j0.2 \approx 13.6 + j1 = 13.64 \text{ angle } (4.21)$$

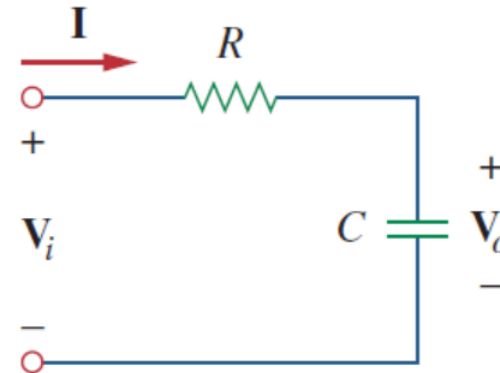
$$I = V/Z_{eq} = 3.67 \text{ angle } (-4.21) \text{ [A]}$$

9.8 Applications

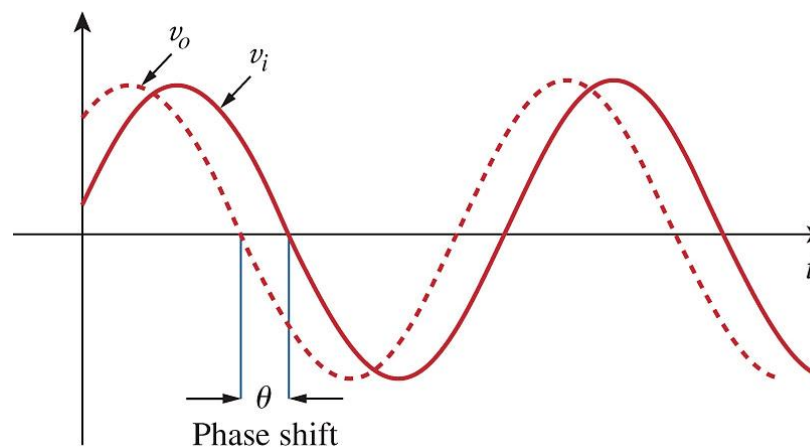
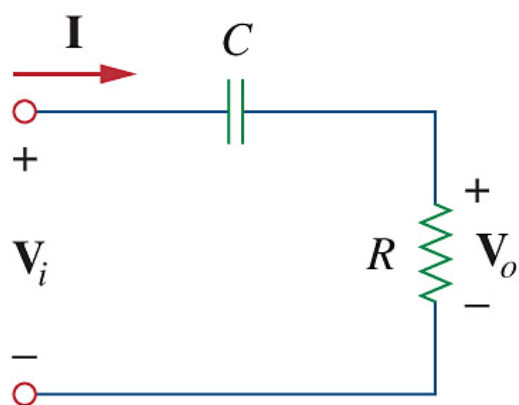
Phase-Shifters: A phase-shifting circuit is often employed to correct an undesirable phase shift already present in a circuit or to produce special desired effects.



V_o leads V_i



V_o lags V_i

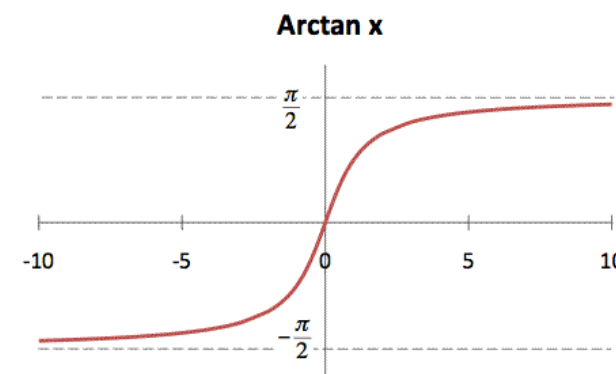


$$\tilde{V}_o = \tilde{V}_i \frac{R}{R + 1/(j\omega C)} = \tilde{V}_i \frac{R}{R - j(1/\omega C)}$$

$$= \tilde{V}_i \frac{R}{\sqrt{R^2 + (1/\omega C)^2} \angle -\tan^{-1}(1/(\omega RC))}$$

$\tilde{V}_o$ leads $\tilde{V}_i$ by $\theta = \tan^{-1}(1/(\omega RC))$,

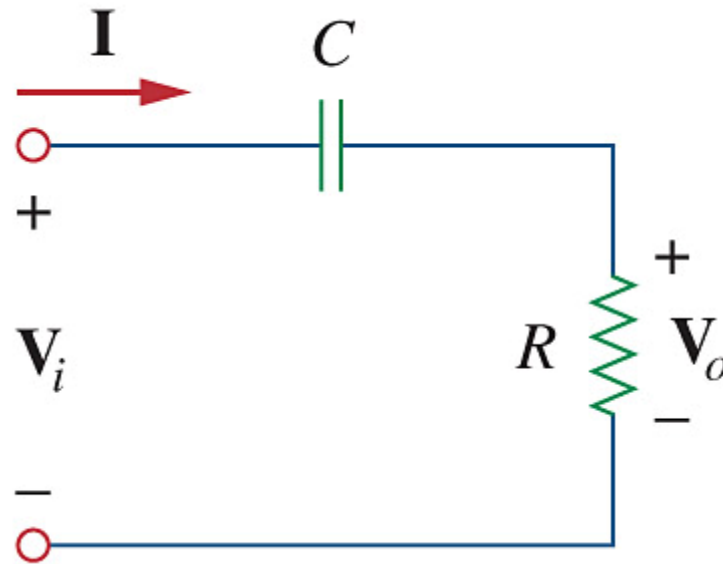
$0^\circ < \theta < 90^\circ$ Amount of phase shift depends on the values of R, C, and ω .



$$\angle \mathbf{V}_i + \theta = \angle \mathbf{V}_o$$

$$\angle \mathbf{V}_o > \angle \mathbf{V}_i$$

$$0^\circ < \theta < 90^\circ$$

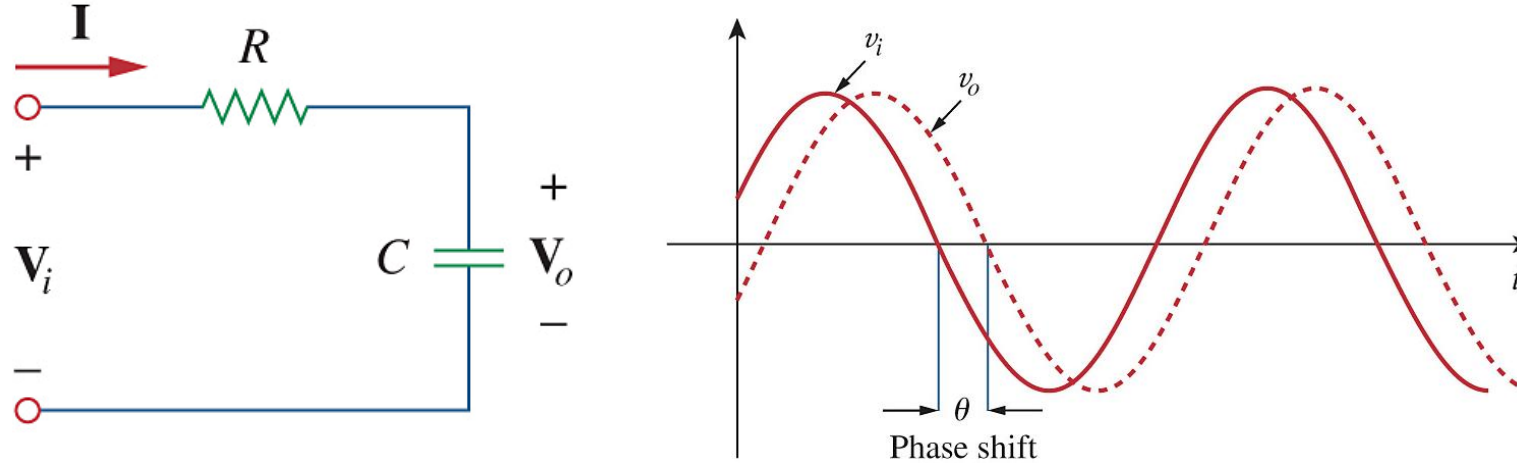


Another way to check leading/lagging relation:

(1) $V_o = IR \rightarrow \angle I = \angle V_o$

(2) $V_i = IZ = I(R + 1/j\omega C) \rightarrow -90^\circ < \angle Z < 0 \rightarrow \angle I > \angle V_i$

(3) Thus, $\angle V_o > \angle V_i$, leading output



$$\begin{aligned}\tilde{V}_o &= \tilde{V}_i \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \tilde{V}_i \frac{1}{1 + j\omega RC} \\ &= \tilde{V}_i \frac{1}{\sqrt{1 + (\omega RC)^2} \angle \tan^{-1}(\omega RC)}\end{aligned}$$

$$\begin{aligned}\angle \mathbf{V}_i - \theta &= \angle \mathbf{V}_o \\ \angle \mathbf{V}_o &< \angle \mathbf{V}_i \\ 0^\circ < \theta &< 90^\circ\end{aligned}$$

$$\tilde{V}_o \text{ lags } \tilde{V}_i \text{ by } \theta = \tan^{-1}(\omega RC), 0^\circ < \theta < 90^\circ$$

Issue of 90° shift:

$\tan\theta$ becomes ∞ when θ approaches 90°

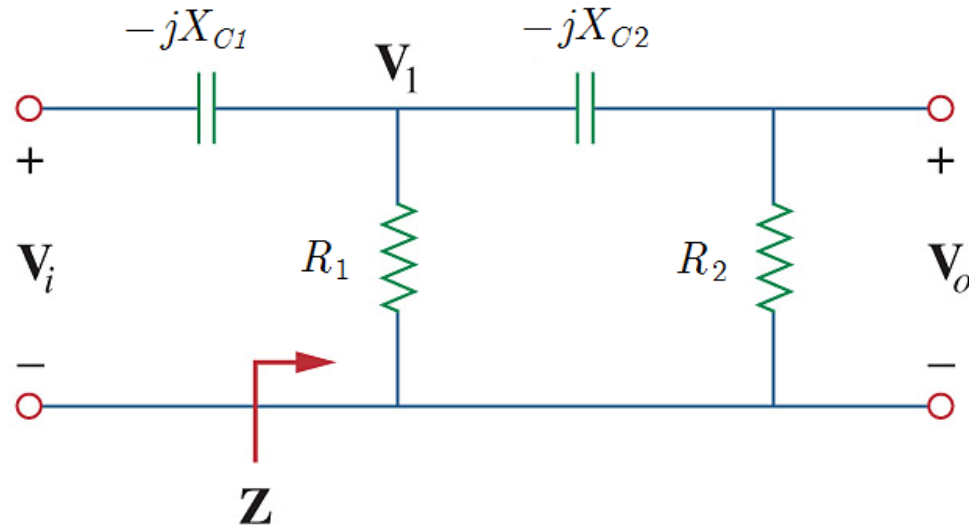
Therefore, $1/\omega RC = \infty$, which means $\omega RC \rightarrow 0$

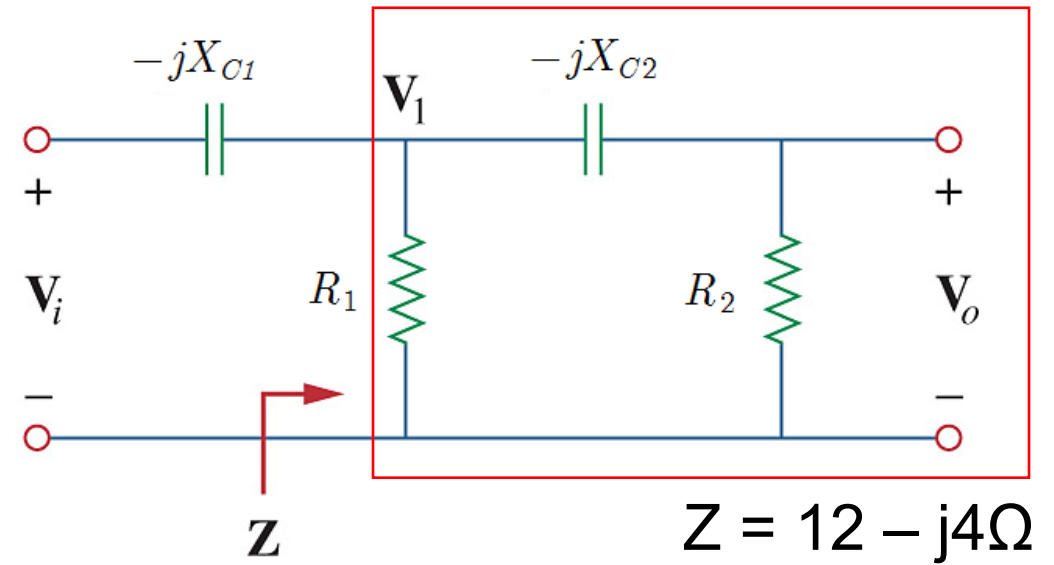
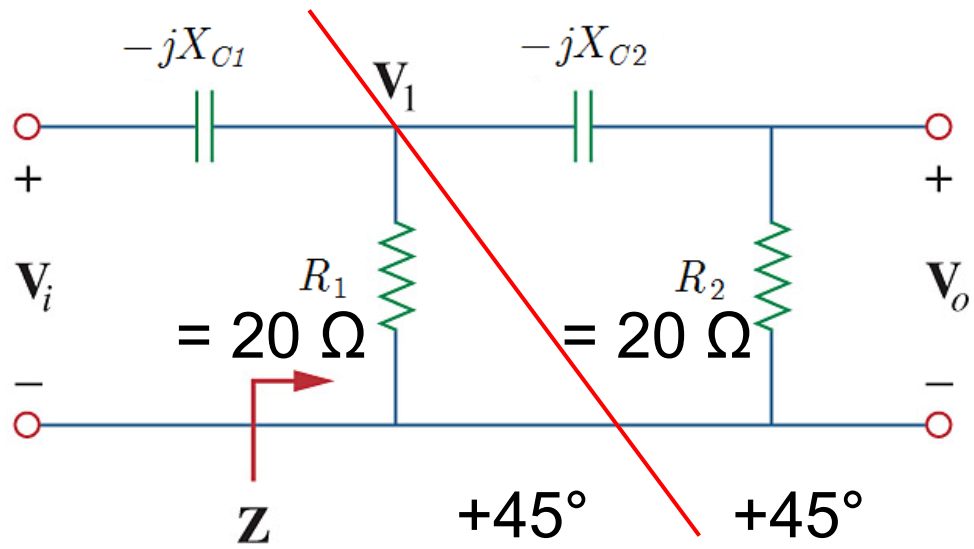
$+1/\omega RC = \tan(+90) = \infty \rightarrow 1/\omega RC = \infty$, i.e. **$\omega RC = 0$**

$$\begin{aligned}\tilde{V}_o &= \tilde{V}_i \frac{R}{R + 1/(j\omega C)} = \tilde{V}_i \frac{R}{R - j(1/\omega C)} \\ &= \frac{1}{1 - j\frac{1}{\omega RC}} \tilde{V}_i = \frac{1}{\sqrt{1^2 + (\frac{1}{\omega RC})^2}} \angle \tan^{-1} \frac{-1}{\omega RC} \tilde{V}_i\end{aligned}$$

No output voltage!

Practice Problem 9.13 Design an RC circuit to provide a phase shift of 90° leading.





We set R_1 and R_2 as $20\ \Omega$

Solution :

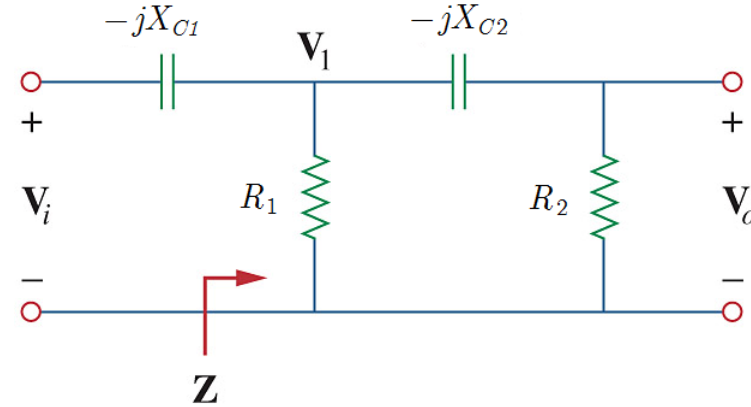
We need two stages, with each stage providing a phase shift of 45° .

Select $R_1 = R_2 = 20 \Omega$,

$$\tilde{V}_o = \tilde{V}_1 \frac{20}{20 - jX_{C2}} = \tilde{V}_1 \frac{20(20 + jX_{C2})}{20^2 + X_{C2}^2}$$

If $X_{C2} = 20 \Omega$, then the second stage produces a 45° phase shift.

$$\begin{aligned} Z &= 20 \parallel (20 - j20) = \frac{20 \times (20 - j20)}{20 + (20 - j20)} \\ &= 12 - j4 (\Omega) \end{aligned}$$



$$\begin{aligned}
\tilde{V}_1 &= \tilde{V}_i \frac{Z}{-jX_{C1} + Z} = \tilde{V}_i \frac{12 - j4}{12 - j(4 + X_{C1})} \\
&= \tilde{V}_i \frac{(12 - j4)(12 + j(4 + X_{C1}))}{12^2 + (4 + X_{C1})^2} \\
&= \tilde{V}_i \frac{(160 + 4X_{C1}) + j(12X_{C1})}{12^2 + (4 + X_{C1})^2}
\end{aligned}$$

For the first stage to produce another 45° ,
we require $160 + 4X_{C1} = 12X_{C1}$, i.e.,
 $X_{C1} = 20 \Omega$.